

EEE 466 (January 2025)

Analog Integrated Circuits and Design Laboratory

Final Project Report

Section: A1 Group: 03

Low-Noise RF Amplifier for Wireless Communication

Course Instructors:

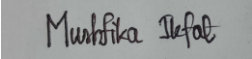
Nafis Sadik, Assistant Professor
Rafid Hassan Palash, Lecturer

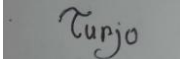
Signature of Instructor: _____

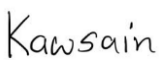
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"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

Signature: 
Full Name: Mushfika Ikfat
Student ID: 2006017

Signature: 
Full Name: Zulkar Naaen Turjo
Student ID: 2006021

Signature: 
Full Name: Kawsain Bin Salim
Student ID: 2006028

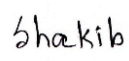
Signature: 
Full Name: Md. Shakib Hossain
Student ID: 2006065

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1 Abstract

This project presents the design and simulation of a high-performance Low-Noise Amplifier (LNA) optimized for operation at 1.8 GHz. The primary objective is to achieve a combination of high gain, low noise figure, and excellent input-output matching while ensuring unconditional stability.

The proposed LNA is implemented using GPD090 CMOS technology, offering a cost-effective and compact solution for RF front-end applications.

The design targets the following key specifications:

- High Gain for improved signal reception sensitivity
- Low Noise Figure (NF) to minimize signal degradation
- Input Matching (S_{11}) and Output Matching (S_{22}) for maximum power transfer
- Stability Factor (K) > 1 to ensure unconditional stability across the operating frequency range

Comprehensive simulations were performed to evaluate the amplifier's performance in terms of S-parameters, noise figure, and stability. The results confirm that the proposed LNA meets the design specifications and is well-suited for integration into modern wireless communication systems.

2 Introduction

In modern wireless communication systems, the Low-Noise Amplifier (LNA) plays a critical role in the receiver front-end by amplifying weak incoming signals while minimizing the addition of noise. This project focuses on the design and simulation of an LNA optimized for operation at **1.8 GHz**, which is a frequency commonly used in mobile and wireless communication applications.

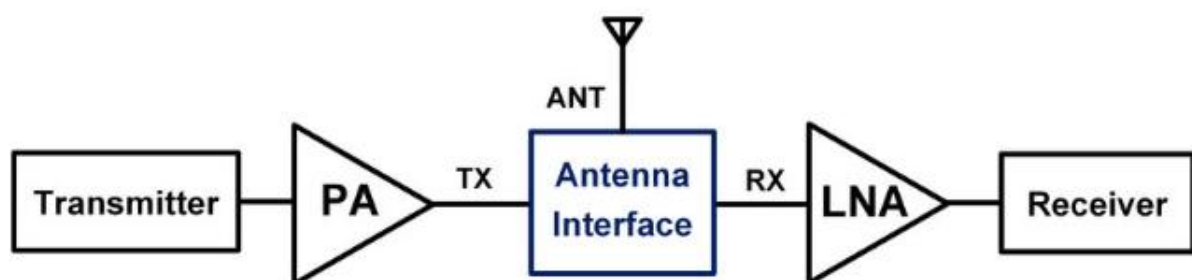


Fig 1: Block Diagram of RF Transceiver Front-End

This is an RF Transceiver Front-End Block Diagram, representing the basic signal flow in a wireless communication system for both transmission (TX) and reception (RX).

1. Transmitter (TX):

Converts the baseband signal into an RF carrier and prepares it for transmission.

2. PA (Power Amplifier):

Increases the power level of the transmitted RF signal so it can be efficiently radiated by the antenna.

3. Antenna Interface (Diplexer / Duplexer / Switch):

Directs the TX signal from the PA to the antenna and routes the received (RX) signal.

4. LNA (Low-Noise Amplifier):

Amplifies weak incoming signals from the antenna while adding minimal noise.

5. Receiver (RX):

Processes the amplified RF signal, down-converts it to baseband, and demodulates it.

The LNA is a key component in the RF signal chain, positioned immediately after the antenna interface. As shown in the system block diagram, it ensures that the desired signal is amplified sufficiently before being processed by the subsequent receiver stages.

3 Design

3.1 Design Specifications

Topology:

Low-Noise Amplifier (LNA)

Specifications:

The goal of this design is to implement a high-performance LNA optimized for operation at 1.8 GHz with the following key performance targets:

- Operating Frequency: 1.8 GHz
- Gain: ≥ 13 dB
- Noise Figure (NF): ≤ 1 dB
- Input Return Loss (S11): ≤ -13 dB

These specifications have been chosen to ensure high signal amplification with minimal noise contribution, as required in RF receiver front-end applications. The design focuses on achieving optimal gain, noise performance, and impedance matching while maintaining stability across the intended frequency band.

3.2 Circuit Diagram

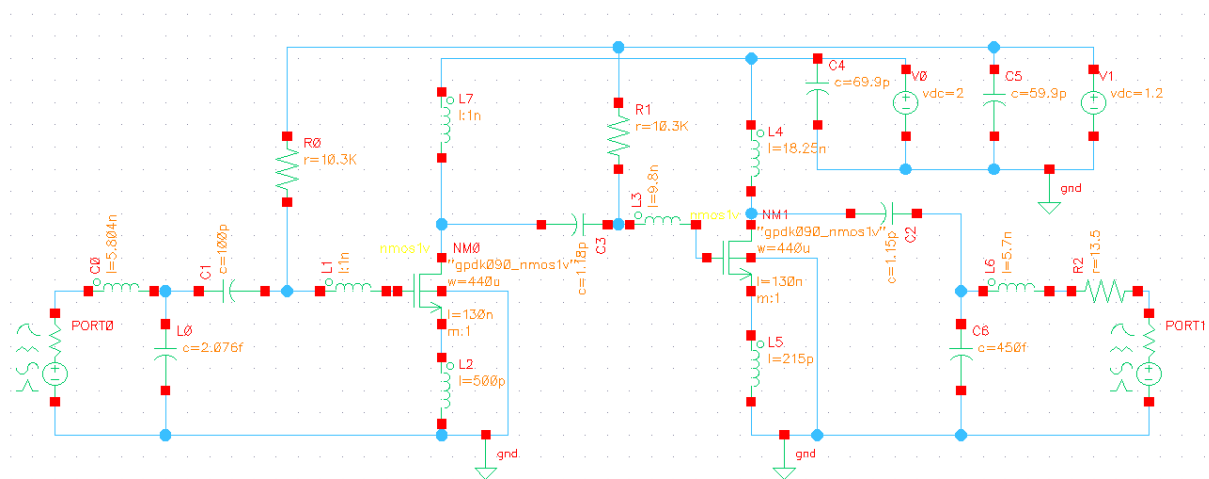


Fig 2: Designed circuit Diagram

This circuit is a two-stage CMOS Low-Noise Amplifier designed for operation at 1.8 GHz. It uses inductive source degeneration for input matching and low NF, along with carefully designed matching networks to achieve the desired gain, stability, and impedance match.

Input Matching Network:

The input Matching Network consists of capacitors C0, C1 and inductor L0, L1 which tunes the input impedance to match $50\ \Omega$ of the port at 1.8 GHz.

First LNA Stage:

Here, the component nmos1v is the main amplifying transistor which has an inductive source degeneration inductor L2 for better input match and low NF. The resistor R0 is connected to the DC voltage source $V_g = 1.2\text{V}$ which provides DC bias for the gate of 1st transistor.

Inter-stage Matching:

The inductor L7, acting as RF choke inductor, blocks RF from flowing into bias supply. The capacitor C3 acts as AC coupling between first and second stage. The resistor R1 is connected to the DC voltage source $V_d = 2\text{V}$ which provides DC bias for the gate of 2nd transistor.

Second LNA Stage:

The component nmos1v is the main amplifying transistor which amplifies signal from first stage for higher gain. The inductors L3 and L4, acting as RF choke inductors, provide biasing while blocking RF to supply. This stage also contains an inductive source degeneration inductor L5.

Output Matching Network:

The output Matching Network consists of capacitors C2, C6, inductor L6 and resistor R2. The capacitor C2 acts as DC blocking capacitor at output of second stage and the capacitor C6 tunes output match for S22. The inductor L6 combining with the capacitors completes the output matching network.

The resistor R2 is specifically used to improve the output reflection co-efficient s22, by the mechanism that the signal while trying to reflect to input is mostly absorbed by this resistor, thereby improving s22.

3.3 Theoretical Description

Input Matching:

Input matching is the process of designing the input impedance of a **Low-Noise Amplifier (LNA)** to match the characteristic impedance of the source, ensuring maximum power transfer and minimal signal reflection.

In most RF systems, the standard characteristic impedance is $50\ \Omega$. A well-designed input matching network minimizes return loss and prevents signal degradation due to impedance mismatch ensuring:

- Efficient power transfer from the antenna to the LNA.
- Minimized signal reflections, improving receiver sensitivity.
- Optimized noise performance, as input matching directly impacts the noise figure of LNA.

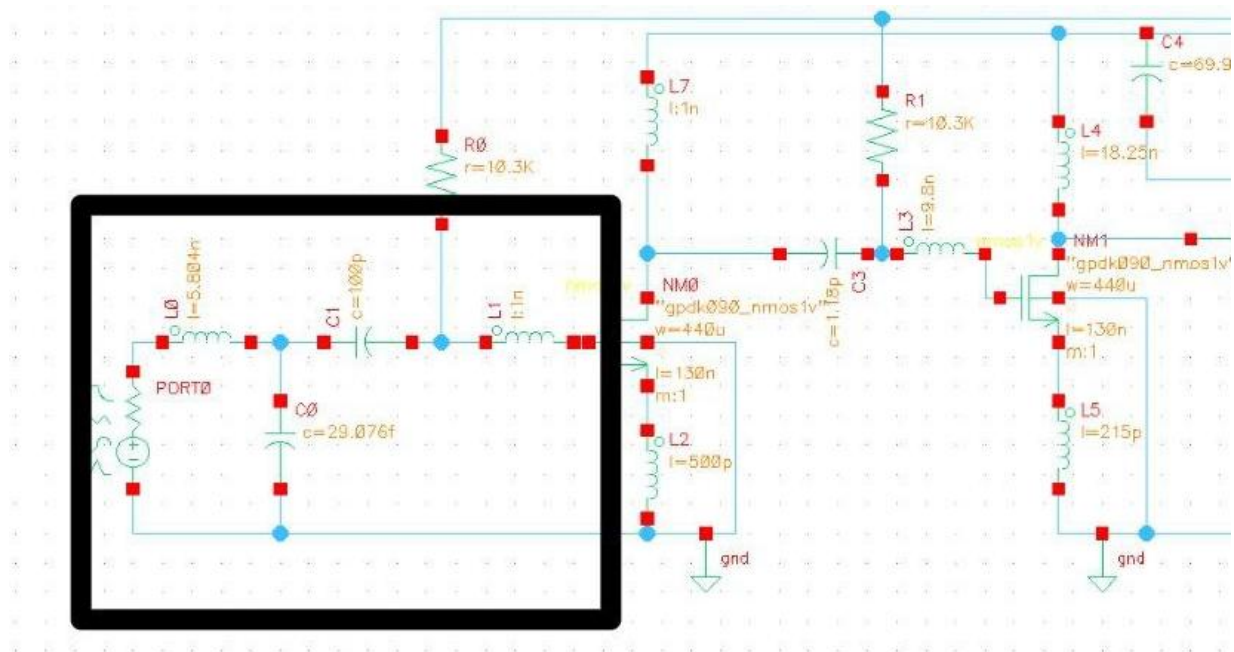


Fig 3: Input Matching Network

In our circuit,

Our input matching network consists of the inductors L0, L1 and capacitors C0, C1. It was designed by adjusting the inductor L0 and capacitor C0 using RF Impedance Matching calculator. The calculator uses the desired operating frequency (1.8 GHz) and the S-parameter characteristics of the device to determine the required matching network values.

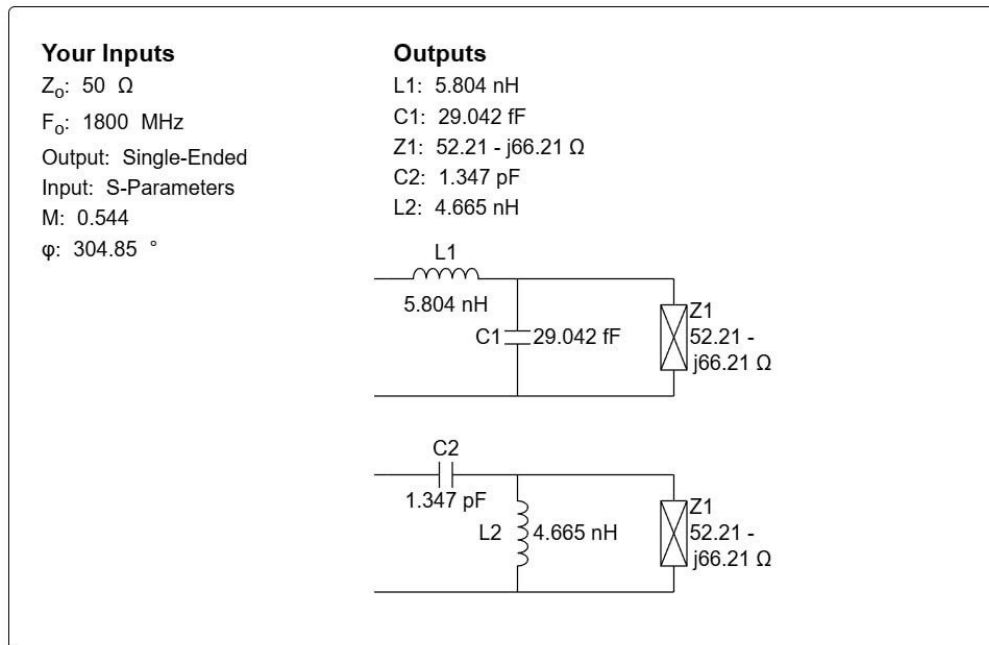


Fig 4: Deriving L0 and C0 values from RF input matching calculator

From the calculation results:

$$L0 = 5.804 \text{ nH}$$

$$C0 = 29.042 \text{ fF}$$

These values provide an input impedance of approximately $52.21 - j66.21 \Omega$, which, when transformed through the network, yields an excellent match to 50Ω at the target frequency.

Output Matching:

Output matching is the process of adjusting the output impedance of the Low-Noise Amplifier (LNA) so that it matches the impedance of the next stage in the RF signal chain. In most RF systems, the standard load impedance is $50\ \Omega$.

Proper output matching ensures that the maximum amount of the amplified signal is transferred to the following stage with minimal loss ensuring:

- Maximize power transfer to the next stage in the receiver chain.
- Minimize signal reflection (reduce return loss at the output).
- Ensure high efficiency and stable operation across the intended frequency band.

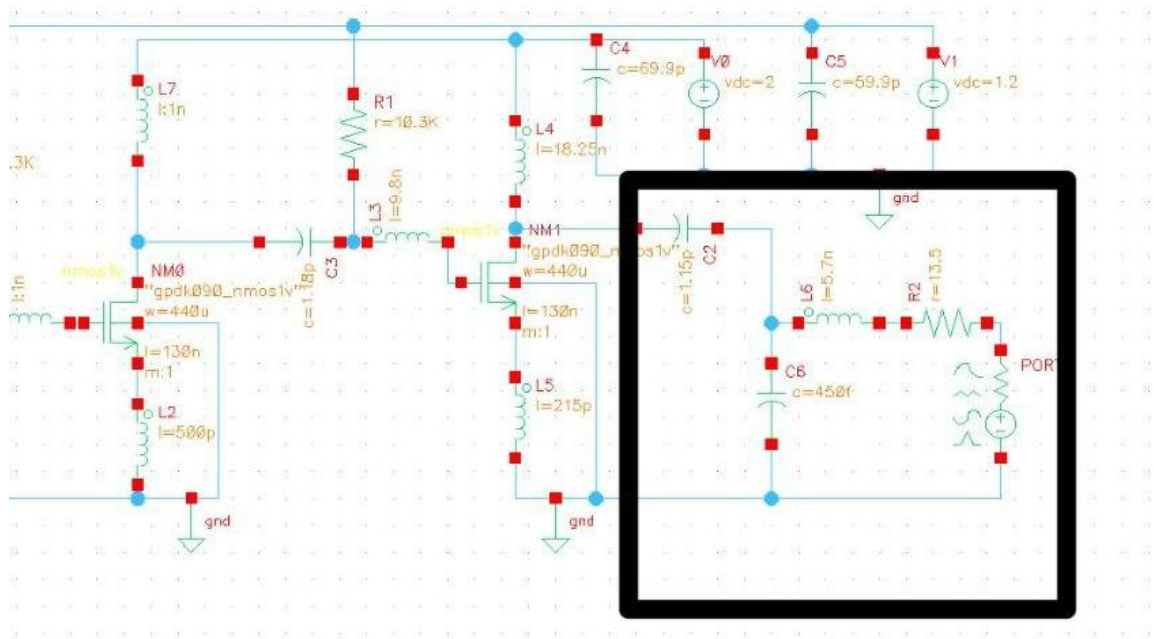


Fig 5: Output Matching Network

In our circuit,

Our output matching network consists of capacitors C2, C6, resistor R2 and inductor L6. It was designed by adjusting the inductor L6 and capacitor C6 using RF Impedance Matching calculator.

These components work together to ensure the output S-parameter (S_{22}) is sufficiently low, indicating minimal signal reflection at the operating frequency of 1.8 GHz. This matching network plays a critical role in maintaining amplifier performance when driving the next stage of the RF receiver chain.

S-parameters:

S11 (Input Reflection Coefficient):

S11 (Scattering Parameter S11) represents the input reflection coefficient of LNA indicating how much of the incident RF signal is reflected back toward the source due to an impedance mismatch at the input port.

Mathematically:

$$S_{11} = \frac{V_{1-}}{V_{1+}}$$

where,

V_{1+} = Incident voltage wave at port 1 (input)

V_{1-} = Reflected voltage wave at port 1

A lower S11 value (more negative in dB) indicates better matching. For this project, the design goal is $S_{11} \leq -13$ dB at 1.8 GHz ensuring efficient coupling of the antenna signal into the amplifier while keeping reflected power low.

S12 (Reverse Transmission Coefficient):

S12 (Scattering Parameter S12) represents the reverse transmission coefficient of LNA measuring how much of the signal from the output port leaks back to the input port. This parameter is a direct indicator of reverse isolation in the amplifier.

Mathematically:

$$S_{12} = \frac{V_{1+}}{V_{2+}}$$

Where:

V_{2+} = Forward incident voltage wave at the output port.

V_{1+} = Voltage wave appearing at the input port due to reverse coupling.

A very low S12 value (e.g., ≤ -20 dB) indicates excellent reverse isolation between output and input, better stability. For this LNA design, the goal is to maintain S12 as low as possible (well below -20 dB at 1.8 GHz) to ensure stable and reliable operation with minimal reverse leakage.

S21 (Forward Transmission Coefficient):

S21 (Scattering Parameter S21) represents the **forward transmission coefficient** or **power gain** of the amplifier. It measures how much of the input RF signal is transferred from the input port to the output port. In LNAs, S21 is a direct measure of **gain**.

Mathematically:

$$S_{21} = \frac{V_{2+}}{V_{1+}}$$

Where:

V_{1+} = Incident voltage wave at the input port.

V_{2+} = Output voltage wave at the output port.

In dB,

$$S_{21(\text{dB})} = 20 \log_{10}\left(\frac{V_{out}}{V_{in}}\right)$$

A higher positive S21 value means more amplification, improving receiver sensitivity, especially important in wireless communication systems where received signals are very weak.

S22 (Output Reflection Coefficient):

S22 (Scattering Parameter S22) represents the output reflection coefficient of LNA indicating how much of the signal at the output port is reflected back toward the amplifier due to an impedance mismatch between the LNA's output and the load (next stage in the RF chain).

Mathematically:

$$S_{22} = \frac{V_{2-}}{V_{2+}}$$

where,

V_{2+} = Incident voltage wave at port 2 (output)

V_{2-} = Reflected voltage wave at port 2

A lower S22 (more negative in dB) means better matching between the amplifier's output and the load. For this project, the design target is S22 is low enough to ensure stable operation and minimal output reflection at 1.8GHz.

Noise Figure (NF):

The Noise Figure (NF) is a measure of how much additional noise an amplifier or system introduces to the signal compared to an ideal, noiseless system. It quantifies the degradation of the Signal-to-Noise Ratio (SNR) as the signal passes through the amplifier.

In decibels:

$$\text{NF(dB)} = 10\log_{10}\left(\frac{\text{SNR}_{in}}{\text{SNR}_{out}}\right)$$

Where:

SNR_{in} = Signal-to-noise ratio at the amplifier input

SNR_{out} = Signal-to-noise ratio at the amplifier output

Lower NF improves receiver sensitivity, allowing detection of weak signals even in noisy environments. Minimizing NF is crucial for front-end amplifiers because noise added at this stage is amplified by all subsequent stages in the receiver chain.

Stability Factor (K):

The Stability Factor is a key parameter in RF and microwave amplifier design used to determine whether the amplifier will remain stable for all possible source and load impedances. For LNAs, stability is essential to prevent unwanted oscillations, which can severely degrade performance.

The stability factor is calculated as:

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12} \cdot S_{21}|}$$

Where:

$$\Delta = S_{11}S_{22} - S_{12}S_{21}$$

When $K > 1$ and $|\Delta| < 1$, the system is Unconditionally Stable. The amplifier will not oscillate under any source or load condition.

When $K < 1$ and $|\Delta| > 1$, the system is Potentially Unstable. The amplifier may oscillate, producing unwanted signals even without input.

4 Simulation

Parametric Analysis:

To improve both input and output matching of the LNA, parametric analysis was performed by sweeping the inductor and capacitor values in the input and output matching networks, two at a time.

This method allowed exploration of how component combinations affect the reflection coefficients S11 and S22. Both S11 and S22 were improved simultaneously, achieving values below the required thresholds. This iterative approach proved effective in balancing the input and output match for optimal performance.

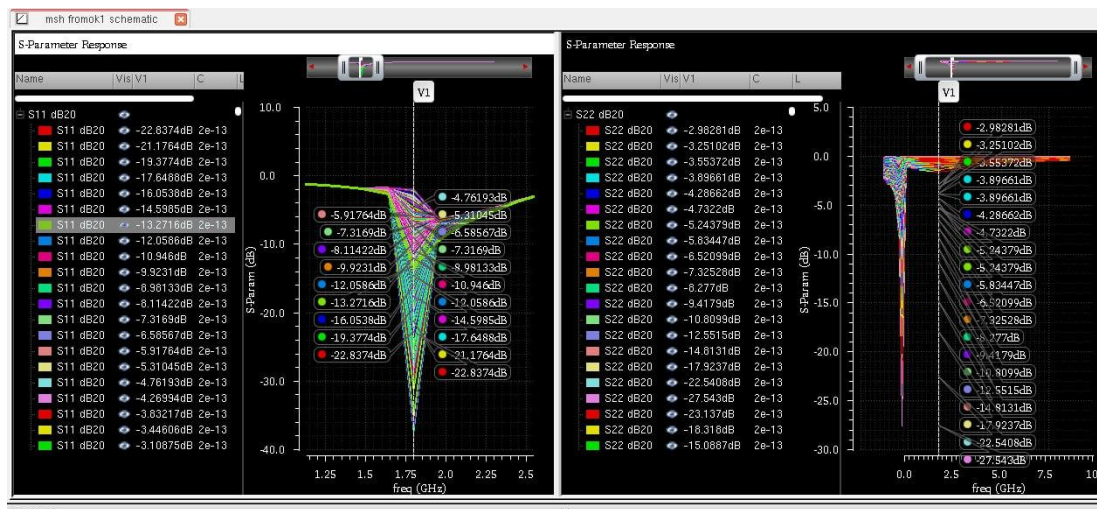


Fig 6: Sample Parametric Sweep

This figure shows an example of a parametric sweep done on input matching network's capacitor and output matching network's inductor and plotted the S11 and S22 together to better search for the optimal point.

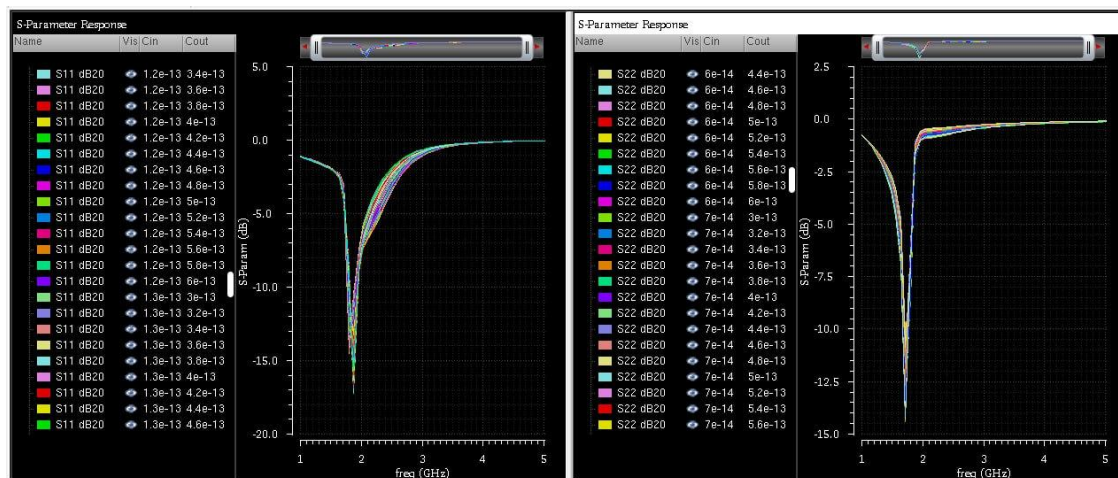


Fig 7: Sample Parametric Sweep

This figure also shows a parametric sweep which was done on input matching network's capacitor and output matching network's capacitor.

Simulation of S11 parameter:

Expected $S_{11} < -13$ dB

Simulated $S_{11} = -14.4699$ dB

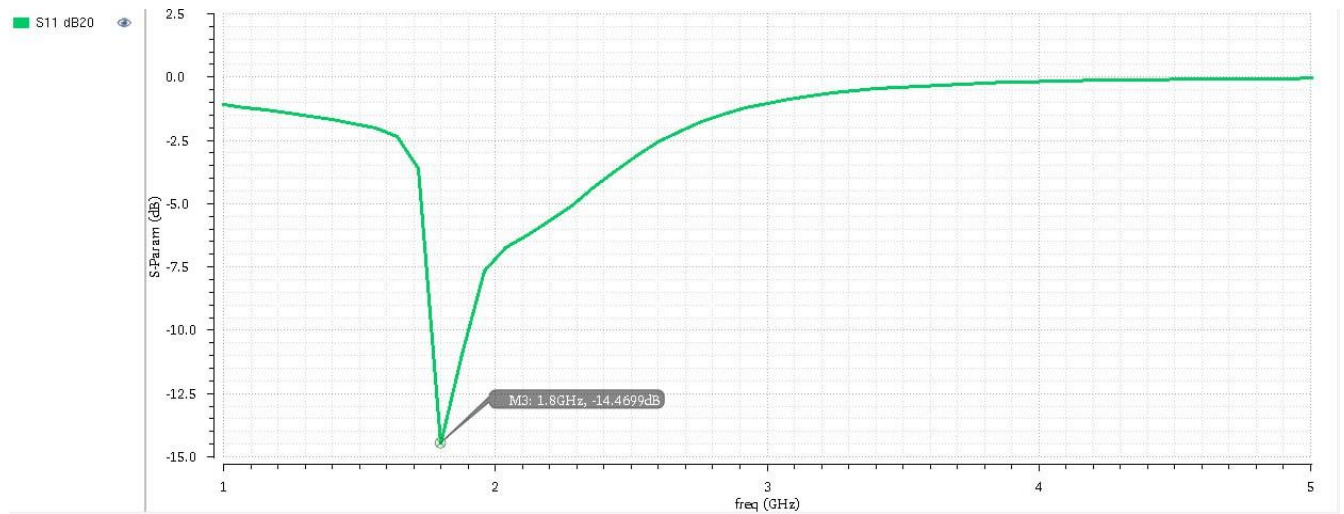


Fig 8: simulation of S11 parameter

The final design achieved $S_{11} = -14.4699$ dB, which not only meets but exceeds the requirement. This means that **less** than 3.6% of the input signal is reflected toward the source, while over 96% is successfully delivered into the amplifier.

Achieving S_{11} well below the threshold indicates strong input matching, reduced signal loss, and improved efficiency in capturing weak incoming RF signals — all of which are critical for high-performance LNA operation in wireless communication systems.

Simulation of S22 parameter:

Expected $S_{22} < -10$ dB to get stability factor > 1

Simulated $S_{22} = -10.0554$ dB

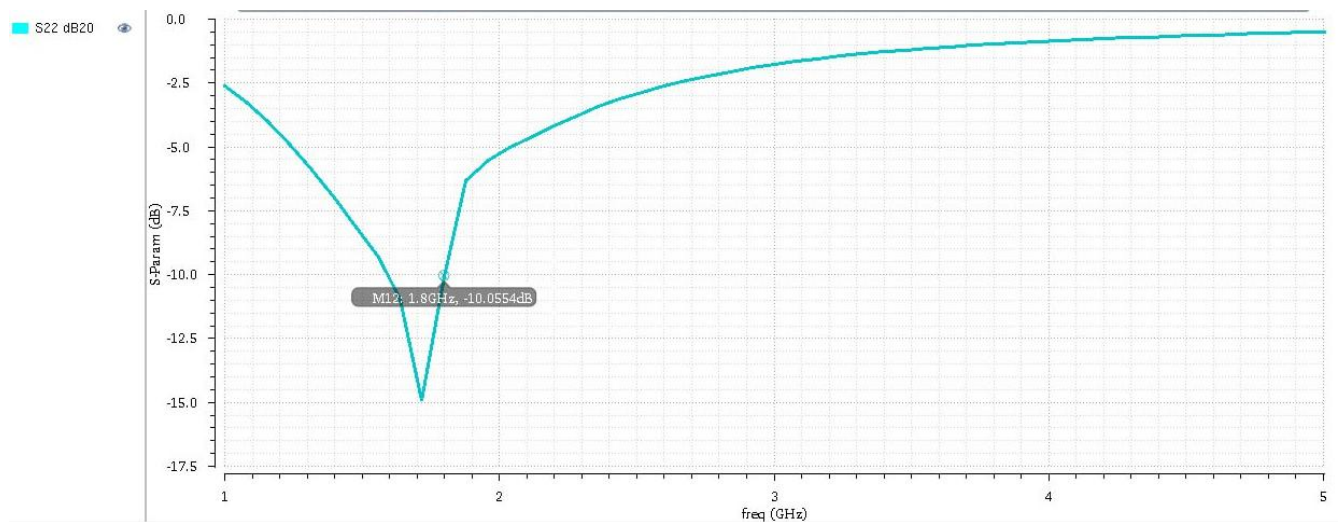


Fig 9: simulation of S22 parameter

In this project, the obtained $S_{22} = -10.054$ dB meets the required condition of being less than -10 dB, meaning the output is well matched to $50\ \Omega$ at the operating frequency of 1.8 GHz. This ensures that the amplified signal is efficiently delivered to the next stage with minimal power loss. Achieving this level of output matching contributes to the overall gain and reliability of the amplifier in a wireless communication system.

Initially, the minimum obtained S_{22} was -8 dB and we got stability factor less than 1. So, we needed to add a resistor in the output before the output port. It reduced the signal reflection and made the output appear closer to $50\ \Omega$ which decreased S_{22} and improved the output matching.

Simulation of S_{21} parameter:

Expected $S_{21} > 13$ dB

Simulated $S_{21} = 34.0766$ dB

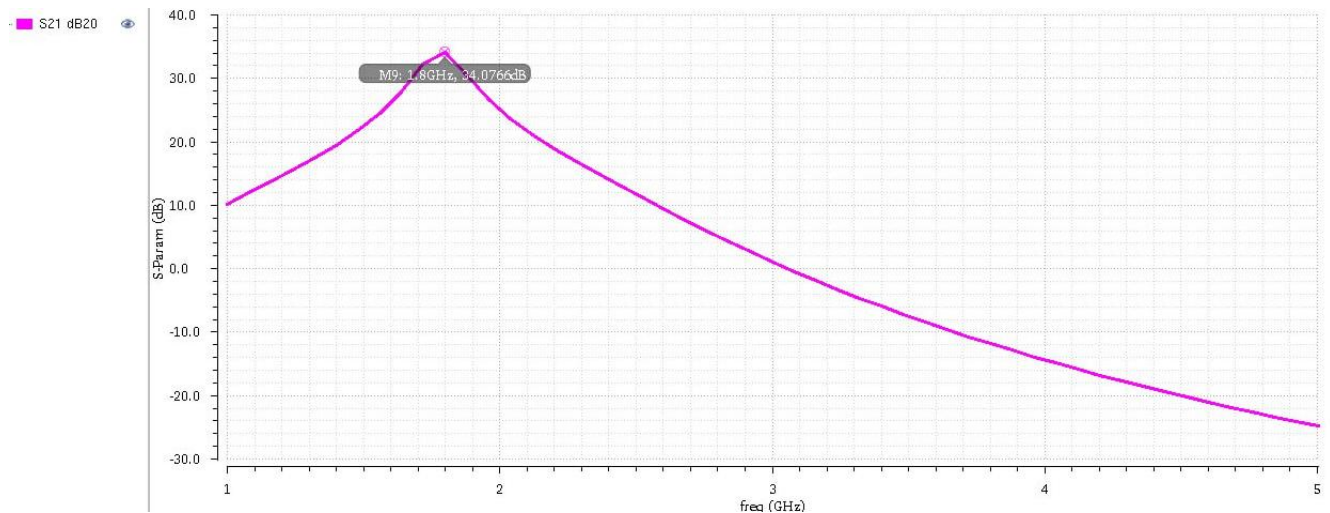


Fig 10: simulation of S_{21} parameter

This high gain indicates that the LNA effectively amplifies weak RF signals while maintaining stability and matching. Such a strong gain ensures that the signal strength is greatly improved before reaching the next stage, which is crucial in low-power or long-range wireless communication systems.

Achieving this level of gain confirms the success of the amplifier design in meeting and surpassing performance expectations.

Simulation of S12 parameter:

Simulated S12 = -39.3075 dB

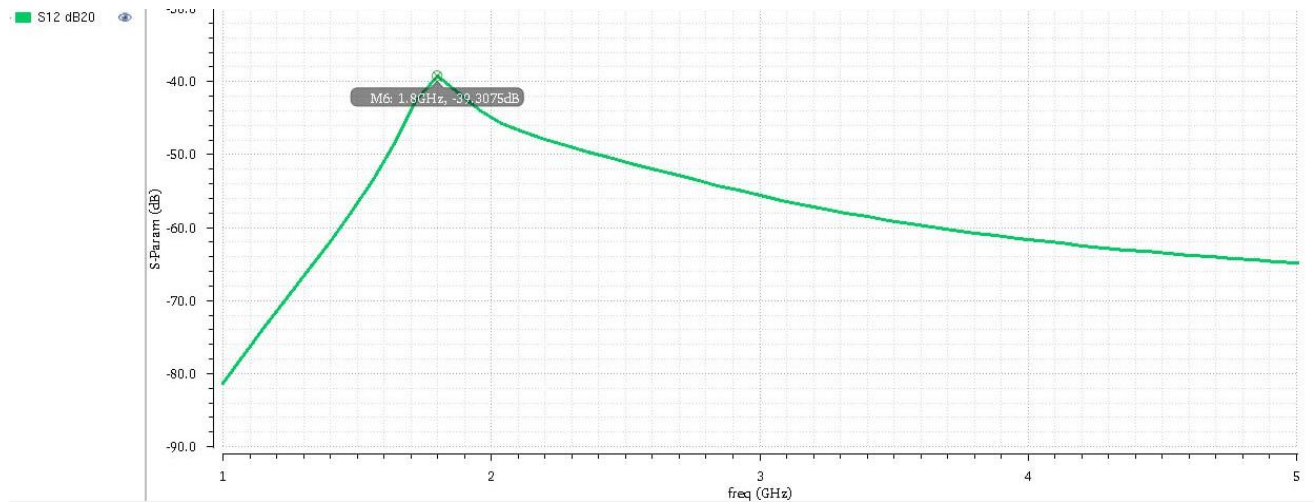


Fig 11: simulation of S12 parameter

A low S12 value indicates that very little of the output signal is fed back to the input, which helps prevent unwanted feedback and oscillations. This high level of isolation is especially important in LNA designs to ensure that the input and output stages remain independent, allowing stable performance and easier impedance matching.

The achieved S12 confirms that the topology used in the design effectively minimizes reverse transmission, contributing to the overall stability and reliability of the amplifier.

Simulation of Noise Figure:

Expected NF < 1 dB

Simulated NF = 0.246826 dB

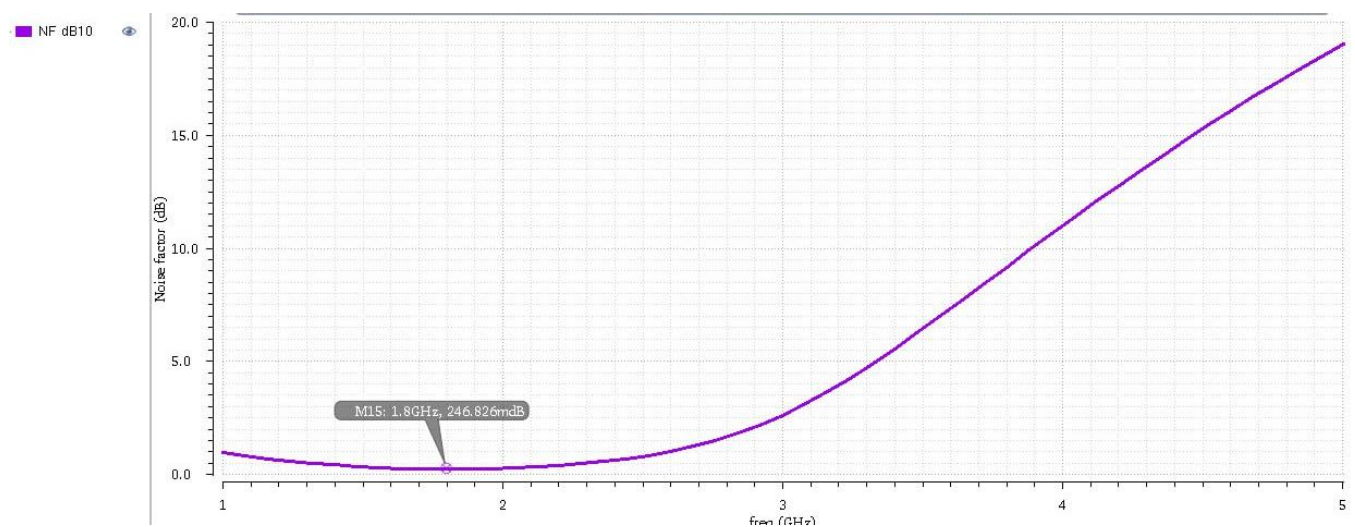


Fig 12: simulation of Noise Figure

The project aimed for a noise figure (NF) of less than 1 dB to ensure low noise contribution from the LNA. The simulated result achieved an excellent NF of 0.2468 dB, which is well below the target. This indicates that the amplifier adds very little noise to the received signal, preserving signal quality and sensitivity, especially important when dealing with weak RF inputs.

Such a low noise figure confirms that the input stage design and biasing were highly effective, making the LNA suitable for high-performance wireless communication systems.

Simulation of Stability Factor k_f :

For unconditional stability k_f should be greater than 1 across the bandwidth. Our simulated value at 1.8 GHz shows $k_f = 1.008726$.

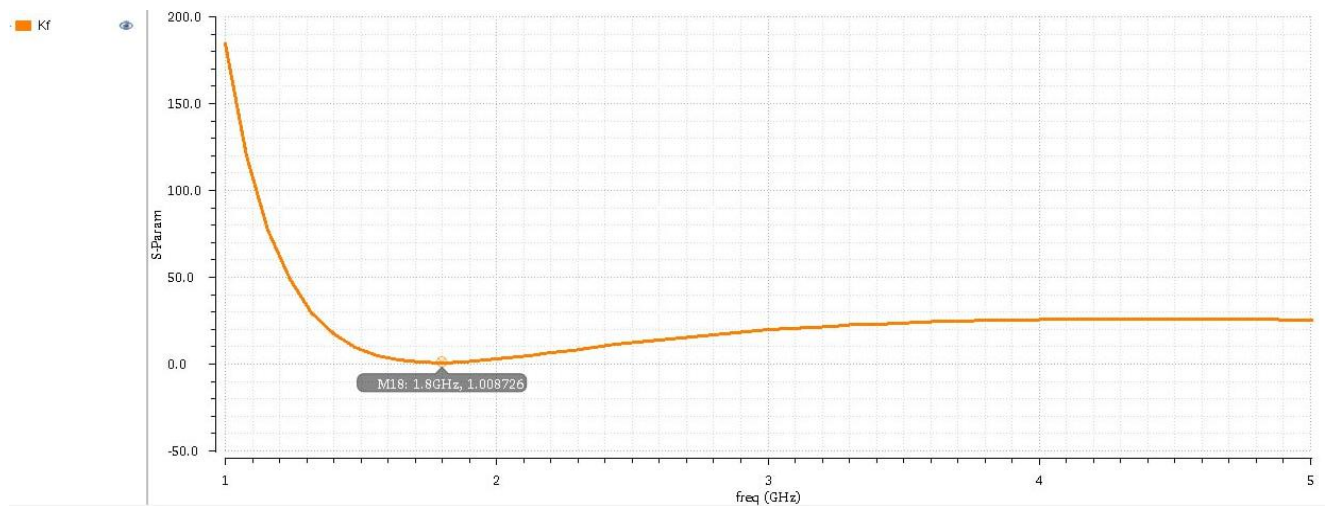


Fig 13: simulation of Stability Factor

And k_f values for other frequencies are also greater than 1. So, our designed LNA is unconditionally stable.

5 Design Analysis and Evaluation

5.1 Novelty

- **Use of Online Matching Calculator:**

During the initial design phase, an online matching calculator was employed to streamline the matching network development process. This tool enabled the rapid calculation and optimization of key matching network components such as inductors and capacitors based on impedance requirements.

- **Technology Platform and Design Environment:**

The proposed circuit was designed and implemented using the gpdk090 (90 nm CMOS) technology, a generic process design kit widely adopted for academic and prototyping applications. The entire schematic capture, simulation, and layout processes were carried out in the Cadence Virtuoso environment, a leading tool for custom IC design.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to public health and safety

The design was developed to operate at 1.8 GHz, a frequency chosen specifically to avoid interference with critical communication and safety-related systems. Additionally, the low-power nature of the RF amplifier ensures reduced electromagnetic (EM) exposure, promoting safer usage in both residential and commercial environments.

5.2.2 Consideration to environment

The development process was entirely simulation-based, resulting in zero physical waste and eliminating the need for prototype fabrication at the early stages. This approach significantly reduces environmental impact compared to conventional trial-and-error hardware prototyping.

Moreover, the resulting amplifier design supports the development of future energy-efficient wireless systems, aligning with global efforts to reduce overall power consumption in communication networks.

5.2.3 Considerations to cultural and societal needs

The amplifier design enables accessible and affordable RF solutions that can be deployed in rural and remote areas, where reliable communication infrastructure is often lacking. This approach fosters innovation while addressing societal needs for connectivity and educational equity.

5.3 Limitations of Tools (PO(e))

Limitations of gpdk090 PDK

The gpdk090 library is a generic PDK, not optimized for actual industrial RF process corners. As a result, it may exhibit reduced accuracy when predicting parasitics or assessing the impact of process variation on performance.

This limitation necessitated cautious interpretation of simulation results, with the understanding that physical implementations may differ slightly from the simulated outcomes.

Tool Constraints in Cadence Virtuoso

Another significant limitation arises from the tool's lack of automated optimization features. Matching network synthesis and bias tuning needed to be performed manually, often requiring multiple iterative adjustments.

Furthermore, Cadence Virtuoso does not provide a built-in matching calculator, which is essential for initial network parameter estimation. To overcome this limitation, we adopted an online matching calculator during the early design stage.

5.4 Impact Assessment (PO(f))

5.4.1 Assessment of Societal and Cultural Issues

Our design promotes accessibility by enabling cost-effective RF solutions for rural and remote areas. Additionally, we used free academic tools (gpdk090), fostering inclusive engineering education and encouraging innovation among students and researchers worldwide.

5.4.2 Assessment of Health and Safety Issues

We designed the amplifier to operate at 1.8 GHz, ensuring it avoids interference with critical systems such as medical and emergency communication devices. The low-power RF design helps minimize electromagnetic (EM) exposure, making it safe for both users and surrounding environments.

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

Name	ID	Contribution
Mushfika Ikfat	2006017	Found key paper and by tweaking on it, met specifications - gain, NF, and near-S11 with two notches at 1.8 GHz
Zulkar Naaen Turjo	2006021	Searched supporting papers; assisted with documentation and analysis, and relevant calculations
Kawsain Bin Salim	2006028	Used online matching calculator for efficient impedance matching, searched relevant papers and built the primary theoretical base for further simulations and tweaking
Md Shakib Hossain	2006065	Optimized S-parameters (S11, S21, S22, S12) and NF through simulation and solved the double notch problem of S11 occurring previously after simulation

6.2 Log Book of Project Implementation

Time Frame	Activities
1 June – 13 June	Literature survey; studied LNA design papers and theory.
14 June – 24 June	Simulated circuits from papers; learned to extract RF parameters in Cadence Virtuoso.
25 June – 7 July	Achieved gain ≥ 13 dB and S11 ≤ -13 dB near 1.8 GHz.
8 July – 18 July	Diagnosed S11 issues; refined matching networks and improved impedance matching.
19 July – 26 July	Met all final specs at 1.8 GHz (gain, NF, S11); finalized simulation results.

7 YouTube Link

Here's the link to our demonstration video:

<https://www.youtube.com/watch?v=tXrC75GiM9c>

8 Future Work (PO(I))

- **Layout Design and Post-Layout Simulation**

The next step involves completing the layout design of the circuit using Cadence Virtuoso. Post-layout simulation will be performed after implementing the physical layout to capture real-world effects such as parasitic introduced by interconnects and device geometries.

Standard checks, including Design Rule Check (DRC) and Layout Versus Schematic (LVS) verification, will ensure compliance with fabrication rules and design correctness.

- **Power Optimization**

Further improvements will focus on power efficiency, particularly by optimizing biasing networks and exploring alternative circuit-level techniques to reduce overall power consumption without compromising gain or linearity. Lower power operation is critical for battery-powered and energy-sensitive applications.

- **Technology Scaling**

The design approach will be adapted to smaller process nodes, such as 65 nm CMOS, to achieve higher levels of integration, lower power consumption, and improved frequency response. Scaling to advanced nodes will also enhance future applicability for next-generation wireless technologies.

9 References

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- Design and analysis of CMOS RF receiver front-end of LNA for wireless applications ☆ Mahesh Mudavath, K Hari Kishore, Azham Hussain, C.S. Boopathi, DOI: <https://doi.org/10.1016/j.micpro.2020.102999>
- RF MICROELECTRONICS Second Edition Behzad Razavi